Humana. Claims Payment Policy

Subject: Laboratory – Hormone Testing

Application: Medicare Advantage Products
Policy number: LAB2023005
Published date: 03/2023
Revision date: 05/2026

Disclaimer: *The intended audience of this medical claims payment policy is healthcare providers who treat Humana members. This policy is made available to provide information on certain Humana claims payment policies. This policy is a guideline only and does not constitute medical advice, guarantee of payment, plan pre-authorization, an explanation of benefits, or a contract. This policy does not govern whether a procedure is covered under a specific member plan or policy, nor is it intended to address every claim situation. Claims may be affected by other factors, such as: state and federal laws and regulations, provider contract terms, and our professional judgment. This policy is subject to change or termination by Humana. Humana has full and final discretionary authority for its interpretation and application. No part of this policy may be reproduced, stored in a retrieval system or transmitted, in any shape or form or by any means, electronic, mechanical, photocopying or otherwise without express written permission from Humana. When printed, this version becomes uncontrolled. For the most current information, always refer to the following website:* ***Humana.com/ClaimPaymentPolicies***

Medicare Advantage Payment Policy

As a Medicare Advantage plan, we have developed payment policies to explain coverage more fully for services. The criteria outlined will apply to Medicare Advantage plans except when the procedure(s) has established criteria through the Original Medicare National Coverage Determinations (NCDs) and Local Coverage Determinations (LCDs) as outlined below.

In addition to the policy, claims payments are subject to other plan requirements for the processing and payment of claims, including, but not limited to, requirements of medical necessity and reasonableness and applicable referral or authorization requirements.

- Measurement of serum free testosterone based on total serum testosterone (CPT code 84403**), sex hormone binding globulin (SHBG) (CPT code 84270) or bioavailable testosterone (CPT code 84410*) **IS REIMBURSABLE** for individuals with any of the following conditions:
  - Acromegaly
  - Acute and chronic kidney failure
  - Benign prostatic hypertrophy
  - Cirrhosis or hepatitis
  - Diabetes mellitus
  - Erectile dysfunction
  - Human immunodeficiency virus (HIV) disease
  - Hyperthyroidism
  - Hypothyroidism
  - Infertility
  - Nephrotic syndrome
  - Obesity
  - Polymorphisms
  - Use of estrogens
  - Use of glucocorticoids, progestins or androgenic steroids

- Testing for serum total testosterone measurements (CPT code 84403**) **IS REIMBURSABLE** at the designated frequency in any of the following situations:
  - For individuals taking enzyme inhibitors for prostate cancer: as needed
  - For individuals receiving testosterone replacement therapy: up to eight tests every four years
  - Gender-dysphoric or gender-incongruent persons: as needed
  - Infertility: as needed
  - Symptomatic individuals being evaluated for polycystic ovary syndrome or functional hypothalamic amenorrhea: as needed

- Measurement of serum dihydrotestosterone (CPT code 82642) **IS REIMBURSABLE** for diagnosing 5-alpha reductase deficiency in individuals with ambiguous genitalia, hypospadias or microphallus.

- Measurement of serum estradiol (CPT code 82670, CPT code 82681) **IS REIMBURSABLE** for the following conditions at the designated frequency:
  - For individuals with gynecomastia: once per lifetime
  - For individuals who are experiencing infertility: as needed

- Testing for serum total testosterone (CPT code 84403**), free testosterone (CPT code 84402) or bioavailable testosterone (CPT code 84410*) **IS NOT REIMBURSABLE** in asymptomatic individuals or in individuals with non-specific symptoms.

Applicable procedures for Medicare Advantage plans

Procedure code	Code description
82642	Dihydrotestosterone (DHT)
82670	Estradiol; total
82681	Estrogens; fractionated
84270	Sex hormone binding globulin (SHBG)
84402	Testosterone; free
84403**	Testosterone; total
84410*	Testosterone; bioavailable, direct measurement (e.g., differential precipitation)

**The payment criterion for 84410 applies to all Medicare jurisdictions, except those states awarded to Noridian Healthcare Solutions, LLC and Palmetto GBA.*
***The payment criterion for 84403 applies to all Medicare jurisdictions, except those states awarded to Palmetto GBA.*

Original Medicare Local Coverage Determinations (LCDs)

Contractor name	Policy title	Policy number	Procedure code
Noridian Healthcare Solutions, LLC	Treatment of Males with Low Testosterone	L36538	84410
Palmetto GBA	Treatment of Males with Low Testosterone	L39086	84403 84410

References

A. Salonia (Chair), C. B., J. Carvalho, G. Corona, T.H. Jones, A. Kadioglu, I. Martinez-Salamanca, S. Minhas (Vice-Chair), E.C. Serefoğlu, P. Verze. (2021). European Association of Urology: Sexual and Reproductive Health. https://uroweb.org/guideline/sexual-and-reproductive-health/

AAP. (2017, 10/02/2017). *Avoid ordering LH and FSH and either estradiol or testosterone for children with pubic hair and/or body odor but no other signs of puberty*. ABIM Foundation. Retrieved 10/19/2018 from http://www.choosingwisely.org/clinician-lists/aap-soen-testing-children-pubic-hair-no-other-signs-puberty/

ACOG. (2018a). ACOG Committee Opinion No. 728: Müllerian Agenesis: Diagnosis, Management, And Treatment. *Obstet Gynecol*, *131*(1), e35-e42. https://doi.org/10.1097/aog.0000000000002458

ACOG. (2018b). ACOG Practice Bulletin No. 194: Polycystic Ovary Syndrome. *Obstet Gynecol*, *131*(6), e157-e171. https://doi.org/10.1097/aog.0000000000002656

ACOG. (2019a). ACOG Committee Opinion No. 773 Summary: The Use of Antimüllerian Hormone in Women Not Seeking Fertility Care. *Obstet Gynecol*, *133*(4), 840-841. https://doi.org/10.1097/aog.0000000000003163

ACOG. (2019b). Screening and Management of the Hyperandrogenic Adolescent: ACOG Committee Opinion, Number 789. *Obstet Gynecol*, *134*(4), e106-e114. https://doi.org/10.1097/aog.0000000000003475

Andersson, C. R., Bergquist, J., Theodorsson, E., & Strom, J. O. (2017). Comparisons between commercial salivary testosterone enzyme-linked immunosorbent assay kits. *Scand J Clin Lab Invest*, *77*(8), 582-586. https://doi.org/10.1080/00365513.2017.1339231

Bhasin, S., Brito, J. P., Cunningham, G. R., Hayes, F. J., Hodis, H. N., Matsumoto, A. M., Snyder, P. J., Swerdloff, R. S., Wu, F. C., & Yialamas, M. A. (2018). Testosterone Therapy in Men With Hypogonadism: An Endocrine Society Clinical Practice Guideline. *J Clin Endocrinol Metab*, *103*(5), 1715-1744. https://doi.org/10.1210/jc.2018-00229

Bhasin, S., Brito, J. P., Cunningham, G. R., Hayes, F. J., Hodis, H. N., Matsumoto, A. M., Snyder, P. J., Swerdloff, R. S., Wu, F. C., & Yialamas, M. A. (2018). Testosterone Therapy in Men With Hypogonadism: An Endocrine Society* Clinical Practice Guideline. *The Journal of Clinical Endocrinology & Metabolism*, *103*(5), 1715-1744. https://doi.org/10.1210/jc.2018-00229

Burger, H. G. (2002). Androgen production in women. *Fertil Steril*, *77 Suppl 4*, S3-5. https://doi.org/10.1016/s0015-0282(02)02985-0

Carnegie, C. (2004). Diagnosis of hypogonadism: clinical assessments and laboratory tests. *Rev Urol*, *6 Suppl 6*(Suppl 6), S3-8.

CDC. (2012). *Total Testosterone*. Centers for Disease Control and Prevention. Retrieved 10/23/2018 from https://www.cdc.gov/nchs/data/nhanes/nhanes_11_12/tst_g_met.pdf

CDC. (2021, 07/01/2021). *HoSt/VDSCP: Certified Participants*. Centers for Disease Control and Prevention. Retrieved 09/28/2021 from https://www.cdc.gov/labstandards/hs_certified_participants.html

Corona, G., Goulis, D. G., Huhtaniemi, I., Zitzmann, M., Toppari, J., Forti, G., Vanderschueren, D., & Wu, F. C. (2020). European Academy of Andrology (EAA) guidelines on investigation, treatment and monitoring of functional hypogonadism in males. *Andrology*, *8*(5), 970-987. https://doi.org/10.1111/andr.12770

Corona, G., Rastrelli, G., Di Pasquale, G., Sforza, A., Mannucci, E., & Maggi, M. (2018). Endogenous Testosterone Levels and Cardiovascular Risk: Meta-Analysis of Observational Studies. *J Sex Med*, *15*(9), 1260-1271. https://doi.org/10.1016/j.jsxm.2018.06.012

Gill-Sharma, M. K. (2018). Testosterone Retention Mechanism in Sertoli Cells: A Biochemical Perspective. *Open Biochem J*, *12*, 103-112. https://doi.org/10.2174/1874091x01812010103

Goldman, A. L., Bhasin, S., Wu, F. C. W., Krishna, M., Matsumoto, A. M., & Jasuja, R. (2017). A Reappraisal of Testosterone's Binding in Circulation: Physiological and Clinical Implications. *Endocr Rev*, *38*(4), 302-324. https://doi.org/10.1210/er.2017-00025

Gordon, C. M., Ackerman, K. E., Berga, S. L., Kaplan, J. R., Mastorakos, G., Misra, M., Murad, M. H., Santoro, N. F., & Warren, M. P. (2017). Functional Hypothalamic Amenorrhea: An Endocrine Society Clinical Practice Guideline. *J Clin Endocrinol Metab*, *102*(5), 1413-1439. https://doi.org/10.1210/jc.2017-00131

Hammond, G. L., & Bocchinfuso, W. P. (1995). Sex hormone-binding globulin/androgen-binding protein: steroid-binding and dimerization domains. *J Steroid Biochem Mol Biol*, *53*(1-6), 543-552.

Hassanabad, M. F., & Fatehi, M. (2018). Androgen Therapy in Male Patients Suffering from Type 2 Diabetes: A Review of Benefits and Risks. *Curr Diabetes Rev*. https://doi.org/10.2174/1573399814666180731125724

Heidenreich, A., Bastian, P. J., Bellmunt, J., Bolla, M., Joniau, S., van der Kwast, T., Mason, M., Matveev, V., Wiegel, T., Zattoni, F., & Mottet, N. (2014). EAU guidelines on prostate cancer. Part II: Treatment of advanced, relapsing, and castration-resistant prostate cancer. *Eur Urol*, *65*(2), 467-479. https://doi.org/10.1016/j.eururo.2013.11.002

Hembree, W. C., Cohen-Kettenis, P. T., Gooren, L., Hannema, S. E., Meyer, W. J., Murad, M. H., Rosenthal, S. M., Safer, J. D., Tangpricha, V., & T'Sjoen, G. G. (2017). Endocrine Treatment of Gender-Dysphoric/Gender-Incongruent Persons: An Endocrine Society Clinical Practice Guideline. *J Clin Endocrinol Metab*, *102*(11), 3869-3903. https://doi.org/10.1210/jc.2017-01658

Hembree, W. C., Cohen-Kettenis, P. T., Gooren, L., Hannema, S. E., Meyer, W. J., Murad, M. H., Rosenthal, S. M., Safer, J. D., Tangpricha, V., & T'Sjoen, G. G. (2018a). CORRIGENDUM FOR "Endocrine Treatment of Gender-Dysphoric/Gender-Incongruent Persons: An Endocrine Society Clinical Practice Guideline". *J Clin Endocrinol Metab*, *103*(7), 2758-2759. https://doi.org/10.1210/jc.2018-01268

Hembree, W. C., Cohen-Kettenis, P. T., Gooren, L., Hannema, S. E., Meyer, W. J., Murad, M. H., Rosenthal, S. M., Safer, J. D., Tangpricha, V., & T'Sjoen, G. G. (2018b). CORRIGENDUM FOR "Endocrine Treatment of Gender-Dysphoric/Gender-Incongruent Persons: An Endocrine Society Clinical Practice Guideline". *J Clin Endocrinol Metab*, *103*(2), 699. https://doi.org/10.1210/jc.2017-02548

Kanakis, G. A., Nordkap, L., Bang, A. K., Calogero, A. E., Bartfai, G., Corona, G., Forti, G., Toppari, J., Goulis, D. G., & Jorgensen, N. (2019). EAA clinical practice guidelines-gynecomastia evaluation and management. *Andrology*, *7*(6), 778-793. https://doi.org/10.1111/andr.12636

Khashchenko, E., Uvarova, E., Vysokikh, M., Ivanets, T., Krechetova, L., Tarasova, N., Sukhanova, I., Mamedova, F., Borovikov, P., Balashov, I., & Sukhikh, G. (2020). The Relevant Hormonal Levels and Diagnostic Features of Polycystic Ovary Syndrome in Adolescents. *J Clin Med*, *9*(6). https://doi.org/10.3390/jcm9061831

Kinter, K. J., & Anekar, A. A. (2020). *Biochemistry, Dihydrotestosterone*. StatPearls Publishing, Treasure Island (FL). http://europepmc.org/abstract/MED/32491566

http://europepmc.org/books/NBK557634

https://www.ncbi.nlm.nih.gov/books/NBK557634

Legro, R. S., Arslanian, S. A., Ehrmann, D. A., Hoeger, K. M., Murad, M. H., Pasquali, R., & Welt, C. K. (2013). Diagnosis and treatment of polycystic ovary syndrome: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab*, *98*(12), 4565-4592. https://doi.org/10.1210/jc.2013-2350

Longcope, C. (1986). Adrenal and gonadal androgen secretion in normal females. *Clin Endocrinol Metab*, *15*(2), 213-228. https://doi.org/10.1016/s0300-595x(86)80021-4

Lynch, K. L. (2016). CLSI C62-A: A New Standard for Clinical Mass Spectrometry. *Clin Chem*, *62*(1), 24-29. https://doi.org/10.1373/clinchem.2015.238626

Mathur, R., & Braunstein, G. D. (2010). Androgen deficiency and therapy in women. *Curr Opin Endocrinol Diabetes Obes*, *17*(4), 342-349. https://doi.org/10.1097/MED.0b013e32833ab083

Meldrum, D. R., Gambone, J. C., Morris, M. A., Esposito, K., Giugliano, D., & Ignarro, L. J. (2012). Lifestyle and metabolic approaches to maximizing erectile and vascular health. *Int J Impot Res*, *24*(2), 61-68. https://doi.org/10.1038/ijir.2011.51

Mezzullo, M., Fazzini, A., Gambineri, A., Di Dalmazi, G., Mazza, R., Pelusi, C., Vicennati, V., Pasquali, R., Pagotto, U., & Fanelli, F. (2017). Parallel diurnal fluctuation of testosterone, androstenedione, dehydroepiandrosterone and 17OHprogesterone as assessed in serum and saliva: validation of a novel liquid chromatography-tandem mass spectrometry method for salivary steroid profiling. *Clin Chem Lab Med*, *55*(9), 1315-1323. https://doi.org/10.1515/cclm-2016-0805

Mohammed, M., Al-Habori, M., Abdullateef, A., & Saif-Ali, R. (2018). Impact of Metabolic Syndrome Factors on Testosterone and SHBG in Type 2 Diabetes Mellitus and Metabolic Syndrome. *J Diabetes Res*, *2018*, 4926789. https://doi.org/10.1155/2018/4926789

Molina-Vega, M., Munoz-Garach, A., Damas-Fuentes, M., Fernandez-Garcia, J. C., & Tinahones, F. J. (2018). Secondary male hypogonadism: A prevalent but overlooked comorbidity of obesity. *Asian J Androl*. https://doi.org/10.4103/aja.aja_44_18

Morales, A., Bebb, R. A., Manjoo, P., Assimakopoulos, P., Axler, J., Collier, C., Elliott, S., Goldenberg, L., Gottesman, I., Grober, E. D., Guyatt, G. H., Holmes, D. T., & Lee, J. C. (2015). Diagnosis and management of testosterone deficiency syndrome in men: clinical practice guideline. *Cmaj*, *187*(18), 1369-1377. https://doi.org/10.1503/cmaj.150033

Mulhall, J. P., Trost, L. W., Brannigan, R. E., Kurtz, E. G., Redmon, J. B., Chiles, K. A., Lightner, D. J., Miner, M. M., Murad, M. H., Nelson, C. J., Platz, E. A., Ramanathan, L. V., & Lewis, R. W. (2018, 2018). *Evaluation and Management of Testosterone Deficiency*. American Urological Association. Retrieved 10/22/2018 from https://www.auanet.org/guidelines/testosterone-deficiency-(2018)

Nassar, G. N., & Leslie, S. W. (2018). Physiology, Testosterone. In *StatPearls*. StatPearls Publishing

StatPearls Publishing LLC.

NCCN. (2021a). *Neuroendocrine and Adrenal Tumors Version 3, 2021*. https://www.nccn.org/professionals/physician_gls/pdf/neuroendocrine.pdf

NCCN. (2021b). *Prostate Cancer, Version 1, 2022*. https://www.nccn.org/professionals/physician_gls/pdf/prostate.pdf

Rochira, V., Antonio, L., & Vanderschueren, D. (2018). EAA clinical guideline on management of bone health in the andrological outpatient clinic. *Andrology*, *6*(2), 272-285. https://doi.org/10.1111/andr.12470

Sachdev, S., Cucchiara, A. J., & Snyder, P. J. (2020). Prostate Specific Antigen Concentrations in Response to Testosterone Treatment of Severely Hypogonadal Men. *Journal of the Endocrine Society*. https://doi.org/10.1210/jendso/bvaa141

Sartorius, G., Ly, L. P., Sikaris, K., McLachlan, R., & Handelsman, D. J. (2009). Predictive accuracy and sources of variability in calculated free testosterone estimates. *Ann Clin Biochem*, *46*(Pt 2), 137-143. https://doi.org/10.1258/acb.2008.008171

Schulster, M., Bernie, A. M., & Ramasamy, R. (2016). The role of estradiol in male reproductive function. *Asian J Androl*, *18*(3), 435-440. https://doi.org/10.4103/1008-682x.173932

Shukla, A., Sharda, B., Bhardwaj, S., Kailash, U., Kalani, R., Satyanarayana, L., & Shrivastava, A. (2018). Association Between Serum Testosterone and Serum PSA Among Men With and Without Partial Androgen Deficiency. *Indian Journal of Clinical Biochemistry*, 1-5. https://link.springer.com/article/10.1007/s12291-018-0785-3

Skinner, R., Mulder, R. L., Kremer, L. C., Hudson, M. M., Constine, L. S., Bardi, E., Boekhout, A., Borgmann-Staudt, A., Brown, M. C., Cohn, R., Dirksen, U., Giwercman, A., Ishiguro, H., Jahnukainen, K., Kenney, L. B., Loonen, J. J., Meacham, L., Neggers, S., Nussey, S., Petersen, C., Shnorhavorian, M., van den Heuvel-Eibrink, M. M., van Santen, H. M., Wallace, W. H., & Green, D. M. (2017). Recommendations for gonadotoxicity surveillance in male childhood, adolescent, and young adult cancer survivors: a report from the International Late Effects of Childhood Cancer Guideline Harmonization Group in collaboration with the PanCareSurFup Consortium. *Lancet Oncol*, *18*(2), e75-e90. https://doi.org/10.1016/s1470-2045(17)30026-8

Snyder, P. J. (2020). *Clinical features and diagnosis of male hypogonadism*. Wolters Kluwer. https://www.uptodate.com/contents/clinical-features-and-diagnosis-of-male-hypogonadism

Stanworth, R. D., & Jones, T. H. (2008). Testosterone for the aging male; current evidence and recommended practice. *Clin Interv Aging*, *3*(1), 25-44.

Star-Weinstock, M., & Dey, S. (2019). Development of a CDC-certified total testosterone assay for adult and pediatric samples using LC–MS/MS. *Clinical Mass Spectrometry*, *13*, 27-35. https://www.sciencedirect.com/science/article/pii/S2376999819300017

Sun, G., Xue, J., Li, L., Li, X., Cui, Y., Qiao, B., Wei, D., & Li, H. (2020). Quantitative determination of human serum testosterone via isotope dilution ultra-performance liquid chromatography tandem mass spectrometry. *Mol Med Rep*, *22*(2), 1576-1582. https://doi.org/10.3892/mmr.2020.11235

Teede, H. J., Misso, M. L., Costello, M. F., Dokras, A., Laven, J., Moran, L., Piltonen, T., & Norman, R. J. (2018). Recommendations from the international evidence-based guideline for the assessment and management of polycystic ovary syndrome. *Hum Reprod*, *33*(9), 1602-1618. https://doi.org/10.1093/humrep/dey256

Travison, T. G., Vesper, H. W., Orwoll, E., Wu, F., Kaufman, J. M., Wang, Y., Lapauw, B., Fiers, T., Matsumoto, A. M., & Bhasin, S. (2017). Harmonized Reference Ranges for Circulating Testosterone Levels in Men of Four Cohort Studies in the United States and Europe. *J Clin Endocrinol Metab*, *102*(4), 1161-1173. https://doi.org/10.1210/jc.2016-2935

van der Veen, A., van Faassen, M., de Jong, W. H. A., van Beek, A. P., Dijck-Brouwer, D. A. J., & Kema, I. P. (2019). Development and validation of a LC-MS/MS method for the establishment of reference intervals and biological variation for five plasma steroid hormones. *Clin Biochem*, *68*, 15-23. https://doi.org/10.1016/j.clinbiochem.2019.03.013

Vermeulen, A., Verdonck, L., & Kaufman, J. M. (1999). A critical evaluation of simple methods for the estimation of free testosterone in serum. *J Clin Endocrinol Metab*, *84*(10), 3666-3672. https://doi.org/10.1210/jcem.84.10.6079

Vesper, H., Botelho, J., & Wang, Y. (2014). Challenges and improvements in testosterone and estradiol testing [Invited Review]. *Asian Journal of Andrology*, *16*(2), 178-184. https://doi.org/10.4103/1008-682x.122338

Viana, A., Jr., Daflon, A. C., Couto, A., Neves, D., de Araujo-Melo, M. H., & Capasso, R. (2017). Nocturnal Hypoxemia is Associated With Low Testosterone Levels in Overweight Males and Older Men With Normal Weight. *J Clin Sleep Med*, *13*(12), 1395-1401. https://doi.org/10.5664/jcsm.6832

Wang, A., Arver, S., Flanagan, J., Gyberg, V., Nasman, P., Ritsinger, V., & Mellbin, L. G. (2018). Dynamics of testosterone levels in patients with newly detected glucose abnormalities and acute myocardial infarction. *Diab Vasc Dis Res*, 1479164118802543. https://doi.org/10.1177/1479164118802543

Welker, K. M., Lassetter, B., Brandes, C. M., Prasad, S., Koop, D. R., & Mehta, P. H. (2016). A comparison of salivary testosterone measurement using immunoassays and tandem mass spectrometry. *Psychoneuroendocrinology*, *71*, 180-188. https://doi.org/10.1016/j.psyneuen.2016.05.022

Yun, Y.-M., Botelho, J. C., Chandler, D. W., Katayev, A., Roberts, W. L., Stanczyk, F. Z., Vesper, H. W., Nakamoto, J. M., Garibaldi, L., Clarke, N. J., & Fitzgerald, R. L. (2012). Performance Criteria for Testosterone Measurements Based on Biological Variation in Adult Males: Recommendations from the Partnership for the Accurate Testing of Hormones. *Clin Chem*, *58*(12), 1703-1710. https://doi.org/10.1373/clinchem.2012.186569

Zakharov, M. N., Bhasin, S., Travison, T. G., Xue, R., Ulloor, J., Vasan, R. S., Carter, E., Wu, F., & Jasuja, R. (2015). A multi-step, dynamic allosteric model of testosterone's binding to sex hormone binding globulin. *Mol Cell Endocrinol*, *399*, 190-200. https://doi.org/10.1016/j.mce.2014.09.001

Zitzmann, M., Nieschlag, E., Traish, A., & Kliesch, S. (2019). Testosterone Treatment in Men with Classical vs. Functional Hypogonadism: A 9-Year Registry. *Journal of the Endocrine Society*, *3*. https://academic.oup.com/jes/article/3/Supplement_1/SUN-222/5484291